

Brandon Li

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EDUCATION

University of California, Los Angeles (UCLA)

Bachelor of Science in Computer Science

Expected June 2028

Los Angeles, CA

EXPERIENCE

Software Development Engineer Internship (Incoming)

Amazon (AWS)

Sep 2026 – Dec 2026

Boston, MA

Software Engineering Intern

Google (YouTube)

June 2026 – Sep 2026

San Bruno, CA

- Added monochrome AV1 support to YouTube's Android app to allow thousands of more videos to be watchable.
- Added 10-bit and 12-bit HDR support to AV1 software decoding to enhance video quality and user experience.

Open Source Contributor

VideoLAN

Jan 2026 – Present

Remote

- Added peak limiter and expander audio filters, refactoring a shared dynamics core from the compressor.
- Integrated ggmml and OpenCV dependencies into the build system (contribs) for on-device model execution.
- Ported legacy OpenCV modules to modern C++ and upgraded the SAM2 filter to SAM3 for text-prompts.
- Cut the final VLC binary size by 5% by adding linker-level dead-code stripping to eliminate unused symbols.

Software Engineering Intern

VideoLAN

June 2025 – Sep 2025

Remote

- Developed C/C++ video filters (real-time visual effects and processing) for VLC Media Player (traffic cone).
- Integrated a SAM2 model module with OpenCV to enable real-time object segmentation for visual analysis.
- Implemented YuNet-based ML face detection for real-time tracking of 100+ faces in surveillance systems.
- Built fast video-to-GIF conversion with 5 dithering algorithms for customizable quality and performance.
- Analyzed the AV1 video-codec specification and studied dav1d decoding algorithms to deepen knowledge.

PROJECTS

yt-media-storage | C++, Assembly/SIMD, FFmpeg, Coding Theory, Compression

Feb 2026 – Present

- Popular C++ tool (890 stars, 110 forks) that encodes local files into YouTube videos for storage.
- Achieved a 100x speed-up using inline Assembly, SIMD intrinsics, and OpenMP parallelization.
- Designed lightweight Qt 6 UI and libsodium encryption, enhancing user experience and security.
- Uses robust CRC/Wirehair error-correction to novelly bypass YouTube's compression algorithm.
- Reached #5 on Hacker News (Y Combinator) within 12 hours, generating thousands of views.
- Includes [technical explanation video](#) with over 3.3M impressions, educating over 270k viewers.

mcav | Java, Spring Boot, TypeScript, CI/CD

June 2020 – Present

- Popular Java framework (130+ stars, 10 forks) for building fast and portable media applications.
- Integrates with VLC/mpv with a documented image-processing pipeline API for easier adoption.
- Uses Spring Boot to stream sub-100ms latency audio to usable TypeScript/Howler.js frontend.
- Configured CI/CD pipeline with TeamCity using Oracle Cloud to automate software releases.
- Engineered native media player via JavaCV that is 5x faster than most pure-Java players.

TECHNICAL SKILLS

Languages: Java, C/C++, OCaml, JavaScript/TypeScript, Assembly

Frameworks & Libraries: Spring, Hibernate, FFmpeg, OpenCV, libVLC, React, Next.js, Tailwind

Technologies: Git, MongoDB, AWS, Google Cloud, Oracle Cloud

Awards: Dean's Honor List